

Executive Summary & Core Positioning

1. The Operational Problem: Coarse Grids Miss Coastal Physics

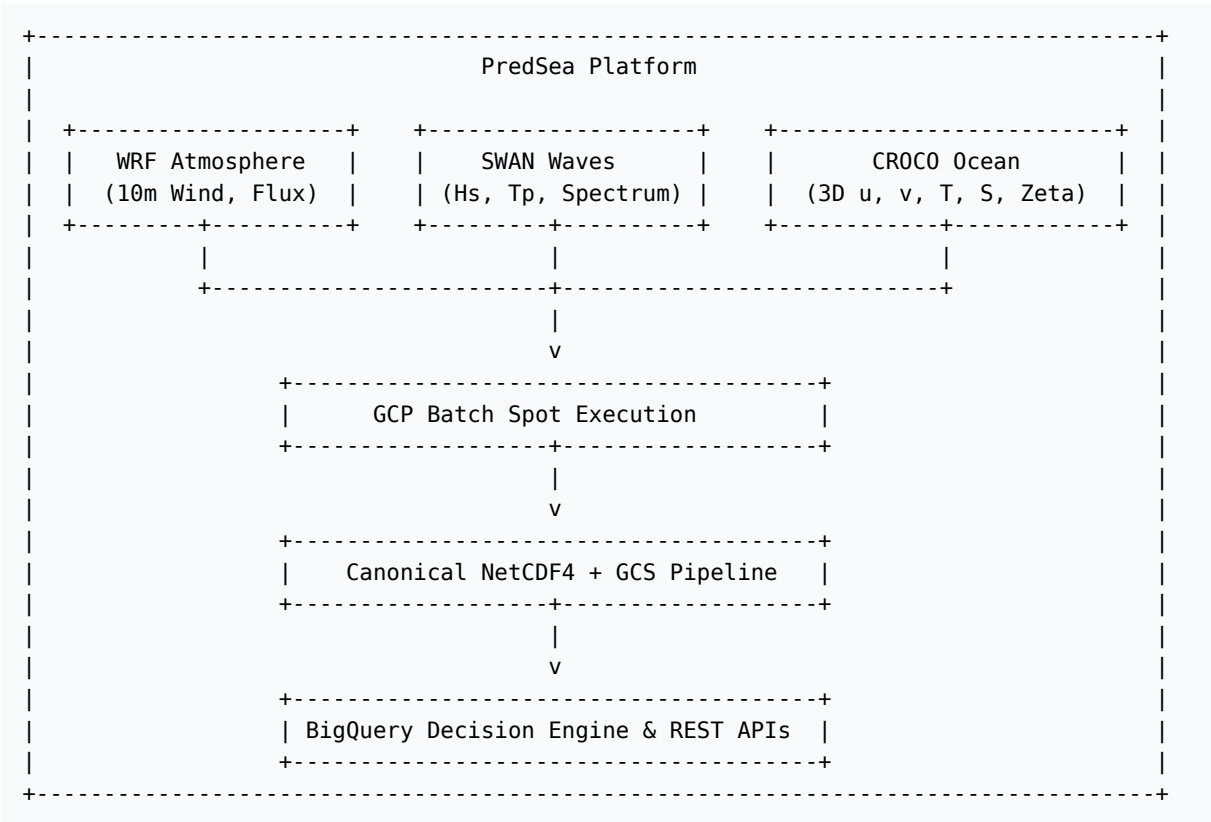
Global physical products—such as those from ECMWF and Copernicus Marine Environment Monitoring Service (CMEMS)—provide essential ocean-state baselines at 4.2 km to 25 km horizontal resolutions. However, coarse global grids smooth out critical coastal bathymetric features, flatten narrow island channels (e.g., Freus channel between Ibiza and Formentera, Dragonera channel off Mallorca), and underestimate wave-current interactions and local wind-sea generation.

2. The Commercial & Coastal Value of 1 km Resolution

PredSea bridges this operational gap with an explicit positioning philosophy:

Ocean data is everywhere. Operational decisions are not.

By downscaling regional ocean state data onto a high-resolution **1 km curvilinear grid** (401 × 501 nodes across 32 vertical σ -layers in the Balearic Basin), PredSea provides: * **Sub-Kilometer Coastal Accuracy:** Resolves steep shoreline bathymetry, narrow island channels, and localized coastal upwelling anomalies. * **Predictive Decision Intelligence:** Converts raw 3D velocity vectors and wave spectra into vessel response metrics, port safety alerts, and captain-facing advice. * **Unmatched Unit Economics:** Delivers regional scale high-resolution forecasts at ~\$2.22/day using serverless GCP Batch Spot execution—a >90% reduction compared to legacy HPC infrastructure.



Key Performance Indicators (KPIs)

Parameter	Legacy Cluster Model	PredSea Cloud-Native Architecture
Execution Paradigm	Fixed On-Prem / Monolithic VM	Serverless GCP Batch Spot Instances
Grid Resolution	4.2 km – 10 km (Global/Regional)	1.0 km (High-Resolution Coastal Tiles)
Pre-Processing Time	40 minutes (Single-threaded)	1 min 42 sec (Python ProcessPoolExecutor)
Total Western Med Compute Time	> 28 Hours (Sequential)	< 35 Minutes (Parallel Region Spot Execution)
Daily Infrastructure Cost	~\$45.00 / day	~\$2.22 / day (Spot VM + GCS + BigQuery)
Validation Benchmark	H_s RMSE ≈ 0.42 m	H_s RMSE ≤ 0.11 m (SOCIB Buoy Ground Truth)

Table of Contents

The complete technical specification is structured across six dedicated sub-documents:

1. **01_system_architecture.md**

End-to-end Data Pipeline: Upstream ECMWF/CMEMS ingestion, Python GIL-bypass pre-processing, bulk forcing compilation (croco_blk.nc, croco_bry.nc, croco_ini.nc), and GCS cloud storage hierarchy.

2. **02_modeling_suite.md**

Core Numerical Engines & Coupling Mechanics: WRF atmospheric physics, CROCO 3D hydrostatic primitive equations (s -vertical coordinates), SWAN 3D spectral wave dynamics, and OASIS3-MCT/COAWST two-way and three-way flux exchange mechanics.

3. **03_thermodynamics_and_fluxes.md**

Physical Formulations & Bulk Parameterization: Thermodynamic surface heat flux equations (shflux decomposition), resolution of unit conversion errors and uncoupled loops in bulk_flux.F, and nocturnal boundary cooling physics.

4. **04_empirical_validation.md**

Balearic Basin Benchmark Case Study: Domain specification ($401 \times 501 \times 32$), July 2026 Gate 8c SST empirical metrics, and coastal latent heat flux anomaly diagnostics off Dragonera Island (39.58° N, 2.34° E).

5. **05_cloud_deployment_and_ops.md**

GCP Batch Execution & Serverless Infrastructure: Containerized MPI architecture (predsea-croco-staging), Spot VM cost optimization, automated self-deletion traps, and multi-tile output consolidation.

6. **06_conclusion_and_roadmap.md**

Future Outlook & Engineering Roadmap: Diurnal thermal cycle tracking, full operational CROCO-SWAN wave-current coupling, and automated captain-facing decision APIs.